

How emergent gameplay in Minecraft brings its community closer

Birk Lerviks
University of Gothenburg
guslerbi@student.gu.se

Kyle Agius
University of Gothenburg
gusagiky@student.gu.se

1 Abstract

In this paper, we explore how emergent gameplay in Minecraft impacts the discourse and communities surrounding the game. We found four key areas where emergent gameplay shapes player interactions: (1) knowledge sharing and specialised subcommunities, (2) custom game modes, maps, and servers, (3) storytelling and content creation, and (4) the continuous evolution of Minecraft. Using a modified version of the MDA framework, we analyse various Minecraft communities from the perspective of mechanics, dynamics and fellowship aesthetics and compile our findings into a set of game design patterns that encourage communities to form and prosper through emergent gameplay in open-world games like Minecraft. The final set of game design patterns include using modular, abstract creativity-driven mechanics; designing complex interactive systems; encouraging knowledge sharing through obscure game elements; allowing players to create custom challenges; and facilitating emergent storytelling.

2 Introduction

Since its initial release in 2011, Minecraft has become globally renowned for its open-ended gameplay, gigantic procedurally generated worlds, endless creativity, complex emergent systems and the communities that have grown around the game. A highly intriguing aspect of Minecraft is how, through its open-ended sandbox mechanics, players are empowered to create, explore and even role play in unexpected and innovative ways, creating a hotbed for emergent gameplay. These player-driven interactions have greatly influenced the discourse and community around Minecraft, and shaped the in-game experience, both in its single-player and multiplayer modes.

This paper aims to identify some of the mechanics and design choices in Minecraft that create the foundations for emergent player-driven gameplay scenarios that are compelling enough to form entire communities around.

Specifically, we aim to answer the following question: *“What are some of the ways that emergent gameplay in Minecraft affects the discourse and community around the game?”*. We used a slightly modified version of the MDA framework (Hunicke et al., 2004), where we analyse different types of Minecraft communities from their Mechanics, Dynamics and Fellowship Aesthetics perspectives. Our usage of the MDA framework is modified in the sense that we are only interested in fellowship aesthetics, while many other types of aesthetics exist. Naturally, our research is both systemic and experiential in nature.

3 Background

This literature review aims to answer how the emergent gameplay systems of Minecraft affect the discourse among players and onlookers, and how communities grow around these systems. As Minecraft is in a state of continuous expansion and development, this review also explores how the game’s communities change the gaming experience, both in how the game is played and how new mechanics and systems are gradually introduced. The results of this research should help developers create systems that better encourage engagement from a community.

Emergent gameplay

(Sicart, 2008) defines game mechanics as the consequences that result from agents, such as a player, interacting with elements in the game. In a complex system, multiple game mechanics and elements affect and react to each other in intricate ways (Tekinbas & Zimmerman, 2003). This is the basis of emergent gameplay.

Emergence in video games can take many shapes, such as game entities interacting to create a working aeroplane, or players building a town hall to discuss in-game politics. Such scenarios are all based on the idea of complex and emergent systems. Though the building blocks, or fundamental

elements of a complex system may be complex themselves, they do not need to be (Tekinbas & Zimmerman, 2003). In fact, complex systems can arise from very simple building blocks. This is often the case in Minecraft (Duncan, 2011), with one prime example being how redstone dust and redstone torches can be linked to create logic gates. By connecting such logic gates in intricate ways, players can simulate boolean arithmetic within the game world (Höltgren et al., 2021).

In emergent systems, new and unpredictable patterns of complexity are created from a limited ruleset, resulting in a whole that is greater than the sum of its parts (Tekinbas & Zimmerman, 2003). In Minecraft, the aforementioned boolean arithmetic may be used to construct a redstone contraption, or machine, that simulates a tiny computer which the player can use to play a game of pong. As a further layer of emergent gameplay, players may hold a pong tournament on their simulated Minecraft computer.

In their paper on the MDA framework, (Hunicke et al., 2004) include a non-exhaustive list of eight aesthetic goals in video games. Because of Minecraft's open-ended sandbox mechanics, procedurally generated world and multiplayer functionality, it could be argued that the game meets at least seven of these aesthetic goals, namely: Fantasy, Narrative, Challenge, Fellowship, Discovery, Expression and Submission. This aesthetic variety is largely due to the emergent nature of the game's various systems.

Emergence in building

A key property of emergence is that the outcomes of such systems are not directly designed by the developers. (Tekinbas & Zimmerman, 2003) refers to this as a "second order design problem". This can be seen in the building system.

If building in Minecraft used premade assets, such as fixed walls and houses, there would be little room for original and unexpected builds to emerge. By instead using general, modular blocks, players can create builds that are not confined to Minecraft's typical mediaeval setting.

This aspect is heightened due to the low detailed art style that Minecraft uses. The shape and texture of blocks are vague enough to be interpreted as different objects, if under the right context. This can be used to add details or colour gradients to builds, regardless of the original material. Figure 1 shows some examples of creative block

interpretation as wooden stairs have been used to create slanted roofs, and blocks of different hues form a gradient from light to dark along the vertical axis of the houses.



Figure 1 - Example of gradients in Minecraft (Grian, 2021)

In Minecraft, creatures, or "mobs" as they are referred to, drop various items when killed. As some items are obtainable only in this way, players are incentivized to exploit the lives of these mobs by farming them as resources (Bull, 2014.). Manually hunting and killing individual mobs is, however, a time consuming and resource intensive process. Many players therefore opt for building mob farms. These are structures explicitly designed to make mobs spawn en masse, and then be killed, either by the player, or through other means such as entity cramming, or fall damage. As mob farms rely heavily on emergent behaviour between various different underlying game systems, they can be quite difficult to design. This has prompted successful mob farm engineers to proudly showcase their designs on the internet, connecting with other like minded people and creating communities where players can improve on each other's contraptions. This process of natural community formation is prevalent across many of Minecraft's systems, with the aforementioned redstone system being another example.

Emergent Storytelling

Emergent stories are narratives that form naturally through the gameplay systems, instead of as part of a pre-written story (Tekinbas & Zimmerman, 2003) (Game Maker's Toolkit, 2018; The Game Overanalyser, 2019). This results in a story that is unique to the player, and allows for more replayability than a fixed narrative. Dwarf Fortress and Rimworld are particularly effective examples of emergent storytelling (Adam Millard - The Architect of Games, 2024; Adams, 2019). These games consist of a vast number of interoperating systems that influence the narrative.

As Minecraft has very little in the way of story and lore, most of the storytelling is done through emergence (Life, 2022; Slater N. (2010); Game Developer (2019)). There are some key aspects of Minecraft's design that encourage this. The survival aspect of the game caused various unplanned stories. This is emphasised by the permanence of changes made to the world, acting as reminders of such stories.

In addition, Minecraft open ended nature lets it act as a platform for creating and telling stories (Minecraft, 2016). Games that rely on emergent storytelling need to encourage the player to consider this story, and make them care about the outcomes (Adam Millard - The Architect of Games, 2024). The use of Minecraft for storytelling is explored in more detail in a later section.

Communities in Games

(Tekinbas & Zimmerman, 2003b) considers a community in games to be any instance of a group of people interacting together to play a game. The interactions of a game community is not limited to interactions that occur during the game itself, but around it as well. This definition is used throughout our research

There are some key principles to consider when discussing communities. Affinity spaces are online areas where people discuss a common interest (Pellicone & Ahn, n.d.). Metagaming refers to what occurs outside of the game (Tekinbas & Zimmerman, 2003b). The way that players interact is itself an emergent system, since there are many small actions that can have complex, interconnected and unexpected outcomes (Tekinbas & Zimmerman, 2003b).

Communities in Minecraft

As Minecraft is the most purchased video game of all time (Forbes India, 2024.; BBC, 2023; Hypebeast, 2023), and can appeal to a wide age range and demographics, the Minecraft community is large and diverse. For the purpose of this literature review, focus is placed on the Minecraft community on YouTube and similar video streaming platforms.

Multiplayer

A clear example of communities in Minecraft is that seen in multiplayer mode. This can be seen when playing together with friends, or on survival multiplayer servers (SMPs). Each

server can have its own social contract for what is and is not accepted. Some servers can create and host entirely new games within Minecraft. Anarchy servers, such as 9b9t, are distinct due to the lack of such contracts.

Hermitcraft is an SMP for YouTubers that has been running for twelve years. Initially, the series was primarily about individual members making their own videos. However, as time went on more emphasis was placed on shared experiences, both on the server and outside of it.

Content creation

Another type of community is that of content creators and fans. Players can share their gameplay on platforms such as YouTube. Even if the viewers are not directly playing Minecraft, it would still be considered a community of people who are interacting with a game.

As an example of such a community, the "Life Series" can be examined. While it varies from season to season, the premise behind the series is that each player only has three lives, and the last player to be eliminated is the winner (All Life Series in Order, n.d.; Animagician, 2023). As the story lines are not scripted and player deaths have a strong impact, emergent storytelling is a key feature of the series. The series has a strong fanbase, which can be seen by the quantity and variety of fan art created for it (Life Series Animatics, n.d.; Traffic Zine, n.d.; Traffic Zine, n.d.). Figure 2 displays an example of fan art from the Life Series.



Figure 2 - Fan art of the Life Series, depicting the winners of each season (Everen, 2023)

Knowledge sharing

Sharing of knowledge is a key example of how players help each other both in-game and through online discourse. A common form of knowledge sharing is walkthroughs. This was particularly important during the early days of Minecraft's release, as there were no tutorials or crafting recipe guides. The unintuitive nature of the game encouraged players to learn from others, whether existing friends or online resources, creating a stronger sense of community (amichail, 2013; Razbuten, 2020).

Knowledge sharing can be tied quite clearly with emergence, since this lets players create their own discoveries that can be shared to other players. Taking redstone as an example, players can take circuits or contraptions that others have built and adapt them or improve upon them, hence improving the knowledge base. In online discussions, content creators crediting other creators for using their builds can help smaller creators find an audience, and can result in creators forming groups (Imp And Skizz Podcast, 2022).

Mojang

Mojang also acknowledges the importance of the community. The Minecon events were used to bring together Minecraft fans in person. While these were not possible during the Covid-19 pandemic, they are now being reintroduced. While the mob vote system was controversial (xisumasays, 2023), it was intended to increase the community's involvement. During Minecraft Live and other official events, Mojang references and brings in YouTubers and other key members of the community (Minecraft, 2024).

When updating the game, Mojang releases weekly snapshot versions that let players try out the features and provide feedback (Minecraft Updates - YouTube, n.d.). Due to the emergence in the gameplay, small changes can have lots of implications, which the players can discover. When Mojang added honey blocks, some fans suggested that these blocks should not stick to slime blocks (Mumbo Jumbo, 2019a). Mojang implemented this change, which led to a large quantity of new redstone contraptions being possible (Mumbo Jumbo, 2019b).

4 Analysis

From our literature review, we can identify certain emergent attributes of Minecraft that are particularly relevant when studying the discourse and community around the game. These attributes can be grouped into four categories: (1) knowledge sharing and specialised subcommunities, (2)

custom game modes, maps and servers, (3) storytelling and content creation, and (4) the continuous evolution of Minecraft. While these categories have some overlapping elements, it is useful to consider them separately as it gives deeper insights into how similar phenomena are related through the game's emergent systems

Knowledge sharing and specialised subcommunities

When Minecraft first launched, it did not have any tutorials or any explanation of what the player should be doing. The player was simply dropped in a vast, largely empty world with no instructions or direction. Even core gameplay elements such as crafting recipes could not be found as instructions in the world, prompting players to search online for tutorials and playthroughs (amichail, 2013; Razbuten, 2020). This quickly sparked a community of Minecraft players teaching each other tips and tricks for how to survive and thrive in the vast open world. With time, and further updates to the game, these tips and tricks evolved from simple instructions on how to play the base game, into tutorials on how to build ever more complex creations such as efficient mob farms, buildings of different styles and intricate redstone contraptions. Inspired by blog posts, forums and YouTube videos, players would build upon and improve the designs of others, often sharing their results online. Soon players would find like-minded people on the internet, forming specialised sub communities around various parts of the game.

Due to its open-endedness and emergent gameplay systems, vanilla Minecraft (or the Minecraft base game without any mods) can be played in a huge variety of ways. Some examples include builders, redstone engineers, adventurers, multiplayer survival and speedrunners. These groups have very different priorities and interact with the game's systems to varying degrees. Builders, for instance, focus mainly on creating unique, interesting and beautiful structures of various styles using the game's vast array of differently textured blocks. Of note is that the sides of each block, measured in-game as 1x1 metres, only have 16x16 pixels, contributing to the game's pixelated and blocky look. These low-fidelity graphics allow for creative texture interpretations, encouraging players to use them in interesting ways.

Builders can select a set of blocks that they will use throughout the build. This set is often referred to as a block palette and will vary depending on building style and player preferences (Grian, 2021). There are even sub communities within the larger building community, with some players

focusing solely on a certain style, such as mediaeval castles or futuristic space stations. Minecraft also has a map feature that lets players take a snapshot of a 128x128 block area where each pixel represents an individual block with a unique colour. This makes it possible to create pixel art completely within the game by manipulating the terrain, prompting some players to share their creations online or even sell them in exchange for rare resources or IOUs on multiplayer servers.

Redstone engineers are also interested in Minecraft's building mechanics, but from another perspective. They combine the game's block types with its redstone mechanics. On the small scale, redstone can be used for simple things such as opening a door with a button or powering rails for minecarts to travel on. However, clever players quickly realised that redstone components such as redstone dust, torches and pistons can be connected to simulate boolean arithmetic (Höltgren et al., 2021) and even form simple electrical components. Much like their real-world counterparts, these electrical components, like clocks, input/output devices and logic gates, can be connected to form advanced machines like a fully working CPU. In fact, Minecraft's redstone system is Turing-complete (Höltgren et al., 2021). Other examples of redstone creations include sorting systems, flying machines and entire game modes and minigames built using only redstone logic.

Highly advanced redstone contraptions were never directly imagined by Minecraft's creators but arose through the clever usage of redstone components and the exchange of ideas and designs within the Minecraft redstone community. As the interplay of redstone components is often unintuitive, the community plays a huge role in educating players how redstone works and how it can be used. One great example of this is quasi-connectivity, which is an artefact of an old bug in the redstone system on Minecraft: Java Edition, but is still often used as an integral part of advanced redstone contraptions to this day. Tutorials about quasi-connectivity and an enormous variety of other redstone creations can be found on YouTube and other video sharing platforms. Thumbnails like the one shown in Figure 3 are common in the Minecraft redstone community on YouTube.



Figure 3: Tutorial showing different redstone contraptions (Mumbo, 2021)

While Minecraft is a largely open-ended game with few clear goals, there is a canonical end to the game. The end condition is achieved by travelling to the aptly named End Dimension and killing the Ender Dragon after which the player can jump into a portal to see the game's credits scroll past the screen. Some players have taken completing the game as quickly as possible as their primary challenge, these are known as speedrunners. Speedrunners may also choose other goals than killing the Ender Dragon as their end condition. Since the world of Minecraft is procedurally generated on load by default, players must come up with interesting ways of using the randomly generated world they are given to achieve their goal. This is known as a randomly seeded speedrun and requires an in-depth understanding of the game's underlying mechanics and emergent systems, encouraging speedrunners to discuss their techniques in dedicated communities. It is also possible for speedrunners to play on the same world multiple times, known as a seeded speedrun, prompting a search for the quickest route possible on the most optimal world. As the number of possible Minecraft worlds is practically uncountable, this search is not destined to end anytime soon.

Custom game modes, maps and servers

In Minecraft, players can choose to play by themselves or online with other players. It is common that friends play together on small private servers or through locally hosted LAN. Tasks necessary for survival are often naturally split between players by their playstyles and interests. For instance, one player may venture down into the mines while another creates a farm and a third builds a collective base for the whole group to use. In other words, players can work together to overcome the dangers of survival Minecraft. Players may also choose to create their own houses to express their unique personal styles.

Inevitably, players get bored of working together at some point and often start engaging in pranks and player versus player, or PvP battles. A typical prank is to place a block of TNT under an inconspicuous looking pressure plate in another player's house, leading to a hilarious (or potentially infuriating) situation when they step on the pressure plate and get blown up. This may trigger an in-game fight to the death, with players gearing up and potentially forming groups in a team versus team battle. PvP may also be the entire point of playing together, sometimes with detailed player-created environments as backdrop, or through elaborate redstone or command block powered minigames such as Spleef.

There exists an enormous amount of public Minecraft servers where players can play together with friends and strangers. These servers host a huge variety of game modes, from vanilla survival and creative mode to elaborate server-specific game modes with large teams of dedicated developers. Notable game modes and servers include 2b2t, an "anarchy" server running vanilla Minecraft where players are allowed to do whatever they want, resulting in interesting emergent phenomena such as the absolute devastation of the spawn area to discourage new players from joining, large scale, month long wars and espionage between in-game factions and bounty hunters who track down and kill other players for profit.

Faction servers are built upon vanilla survival, but allow friends to create factions, or groups, with extra functionality such as being invincible to damage dealt by people in the same faction, teleporting to each other and assigning leaders and roles. Commercial minigame servers such as Hypixel host a multitude of PvP and PvE minigames, often with leaderboards, server-specific rewards and even microtransactions. These minigames range in complexity from simple PvP mediaeval-style fighting and Spleef to elaborate game modes such as Hunger Games, Bed Wars (shown in Figure 4) and Build Battles. These can often be played both individually or with a team and often require many hours of not only in-game building, but also modding and custom scripting to create.



Figure 4: Bed Wars on the Hypixel server (Hypixel server for Minecraft, 2022)

Minecraft's movement mechanics and various block types that affect the player's movement in one way or another allows for the creation of interesting parkour maps. For instance, slime blocks can act as trampolines, honey blocks can be used to slowly glide down walls, fences or glass panes are thin and thus challenging to jump between and water can be used to absorb all fall damage. Parkour maps are commonplace on multiplayer servers, either casually or as PvP with players competing to complete the course the fastest. Interestingly, first-person parkour gameplay has in recent years gained popularity as background footage in TikTok videos, mainly to grab the viewer's attention. Parkour maps are often shared and downloaded on the internet in dedicated communities.

Communities also exist around other types of player created maps, or "custom maps" as they are often collectively referred to as. Examples include adventure maps and escape rooms, some of which are designed to be played together with friends. These maps often require the game's Adventure mode which inhibits the player from destroying blocks, forcing them to use the tools and tips provided by the map creator. In this setting, Minecraft can be considered a game/level design tool or even a complete, albeit limited, game development environment.

Storytelling and content creation

The canonical story of Minecraft is very vague with minimal to no backstory for the game's characters, landscapes and pre-generated structures. Instead, the stories of Minecraft come from the experiences of its players. The perils of Minecraft's survival mode, for instance, are a breeding ground for interesting gameplay scenarios, even in single-player mode (Game Developer, 2019; Life, 2022). Seasoned Minecraft players are likely familiar with their newly built home being decimated by a Creeper explosion,

or falling to their death after mining rare resources in a deep ravine. Such stories arise out of a combination of player motivation and the emergent nature of the game. Take the ravine example; the player needs to acquire diamonds in order to create better armour. When digging down in search of the precious stone, they stumble into a procedurally generated underground ravine which exposes many underground resources and sure enough, diamonds at the very bottom. They venture all the way down and collect the diamonds but upon returning back to the top of the ravine, they are suddenly shot by a skeleton lurking in the darkness, and fall to their death. This gameplay story was not explicitly planned by the developers and is instead a result of the game's procedural systems interacting with each other, guided by the player's motivation. Players share stories like these, making for engaging videos and live stream content.

The open-endedness of Minecraft makes it a great tool for storytelling. As explained before, narratives can be created on-the-go through the game's mechanics; they can, however, also be deliberately handcrafted by content creators. Many player-created stories and narrative settings have elements of both. A prominent example are Minecraft survival multiplayer servers, or SMPs, specifically for YouTubers and streamers. These are servers where all, or most players are content creators, and they record themselves playing together to create interesting and engaging Minecraft videos. This usually yields multiple sub communities within the same SMP, as the creators are uploading videos to their individual YouTube accounts. Viewers may, however, engage with the content of multiple YouTubers of the same SMP and consider themselves to be a viewer of the SMP at large, and not necessarily a particular content creator. Examples of such SMPs are Hermitcraft, SciCraft and the Life Series.

Minecraft SMPs often have a distinct identity and social contract and may have additional rules, or mods on top of the base game. For instance, Hermitcraft is a relatively low-stakes and very inclusive server where creators with wildly different playstyles come together and interact with each other, often through pranks, minigames and server-wide events spanning weeks or months (Hermitcraft Wiki, 2024). The server has also previously utilised a datapack (among others) that allows armour stands to be placed in interesting ways, enabling artistic constructions that are not possible in the base game. In the Life Series, players are pitted against each other in a dramatic game of last man standing, where each player has only three lives before being eliminated from the season. These stakes create interesting dynamics between players and onlookers. SciCraft on the other hand is entirely focused on creating mind boggling redstone contraptions that stretch the boundaries of what is possible in vanilla Minecraft. SMPs like these rely on the game's various systems, broad block

palette, moddability and the interests and personalities of their content creators to create engaging content with identity and integrity.

The continuous evolution of Minecraft

Mojang, the developers of Minecraft, acknowledge the importance of the Minecraft community. Various subcommunities and Minecraft content creators are often mentioned and directly involved in official Minecraft events like Minecon. Mojang has also directly involved the game's fans in the development and direction of the game through events like the hugely popular annual mob vote where players could vote on which new mob they would like to see in the game in the future. This event garnered a lot of interest from players but was also controversial for many reasons, not least of which was that fans felt like Mojang was too lazy for not implementing both mob options in the game (xisumasays, 2023).

As Minecraft is continuously being developed, the studio releases weekly snapshots with the latest updates. There exists a large community around testing the latest snapshot and giving feedback to Mojang. This community of playtesters are invaluable for the game's development as they are often the first outside the studio to spot game-breaking changes that arise from unforeseen interactions between the game's various systems. Mojang also listens to and implements popular ideas from these playtesters, such as when many suggested that honey blocks and slime blocks should not stick to each other which would allow for a whole new set of incredible redstone contraptions and flying machines.

5 Discussion

Based on the previous section, the different aspects of Minecraft's gameplay have been analysed using the MDA framework. This will focus on the link between emergent dynamics and fellowship aesthetics. From these, a set of game design patterns has been created.

Mechanics - Building
The game world consists of large blocks. Blocks can also be placed by the player
There are almost no restrictions regarding where blocks can be placed
Minecraft has a large variety of blocks (around 800), each

with different textures
Elaborate game modes with their own unique mechanics
Maps show a 128x128 block area, with each block type being shown as a solid pixel
Emergent Dynamics
Players can build new structures by placing various blocks. These structures are not predefined and have endless possibilities
Blocks can be interpreted in different ways, depending on the context. Players can use the variety of blocks to add interesting shapes and textures to their builds
Players can create pixel art by manipulating the terrain. This requires a large amount of time and effort
Fellowship Aesthetics
The hidden complexity in building makes it a skill that players can get better at. In a multiplayer setting, this means that some players can take on the role of builder if they are particularly skilled in it
The limitless amount of buildings that can be made gives each community a distinct identity, that helps it feel different and personal. A broad block palette encourages distinguished building styles
Both buildings and building techniques can be shared and discussed online, which gives a sense of fellowship and community even in single player. Players can develop a recognisable building style
Game Design Patterns
Self expression and creativity can be encouraged by using unrestricted, modular components that are open to interpretation
Systems with hidden complexity and skill allow for mastery and specialisation, leading to players taking separate roles

Mechanics - Redstone
Some blocks can produce a redstone signal, with an associated strength
Redstone dust can transfer a signal over distances
Some blocks react to being powered, with different outcomes based on the strength
Emergent Dynamics
Players can make small, reusable circuits. This includes logic gates, timers, et cetera
Players can use these blocks to make working contraptions in the game, such as sorting systems and flying machines

Players were able to build a working, turing-complete computer using the redstone system
Fellowship Aesthetics
As redstone can be rather complex, players can teach other players how it works by creating online tutorials or telling friends. This enforces the importance of the community
Players can improve the contraptions made by other players. This makes the community feel closer, and even small online creators with good designs can have an impact on the community at large
Game Design Patterns
Complex systems can be achieved by having multiple components that interact in different ways. These can lead to players making and sharing their own discoveries.
Leaving some of the information in a game obscure can encourage players seeking out information from other players

Mechanics - Multiplayer
Servers and LAN enable multiplayer
PvP and PvE fighting
Redstone and command block powered minigames
Elaborate game modes with their own unique mechanics
The player's movement is straightforward but must be mastered for advanced manoeuvres
Adventure mode inhibits players from destroying blocks
Many block types affect movement
Emergent Dynamics
Tasks are dynamically created based on the needs of survival
Blocks and items can be used in unexpected ways to achieve a goal
Players can create platforming challenges (Parkour) based on the blocks available and the movement system
Fellowship Aesthetics
Competing in minigames, either as individuals or in teams
Players compete in minigames for the best position on leaderboards, some servers thus act as arcades
Player-created custom maps are shared on the internet
Various items and gear can be used creatively in social settings such as pranks and PvP
Large scale, month long wars and espionage between player groups

Tasks are naturally split between friends by their interests
Game Design Patterns
Letting players create their own challenges can result in a lot of new content.
Having multiple gameplay loops and systems that players can interact with as much or little as they want can encourage creative ways of playing together.

Mechanics - Survival
Hostile mobs can kill the player
Valuable resources are placed in areas with more hostile mobs
Upon dying, the player drops all items on the floor, including any armour.
Items despawn after five minutes for better performance
Emergent Dynamics
Players are likely to die when they have valuable resources on them
After death, they then have 5 minutes to collect the dropped items before losing them
As the player no longer has their armour and has to revisit the same dangerous area, gathering the lost resources can be difficult
Fellowship Aesthetics
Simple resource gathering becomes a more intense narrative, with the 5 minute timer causing tension.
These stories can then be shared with others, either online or on multiplayer servers
Game Design Patterns
Distinct emergent narratives are engaging and sharable, particularly if they change the intensity. Systems can be designed to make it more likely for these stories to occur.
Mechanics - Storylines
The game does not force a goal or story on the player
The world can be shaped in any way that the player wants
Players can customise their avatar by drawing their own skin
Emergent Dynamics
Players can create their own stories and characters
These stories can either be scripted beforehand or allow for emergent changes.
Servers can have different characters interacting in a story

Fellowship Aesthetics
Narratives can be shared online, often as a YouTube series. Viewers can experience this without playing the game.
Game Design Patterns
Encouraging self expression and player freedom allows players to tell and share their own stories. Communities can form around these stories, with less focus being placed on the game itself

6 Future Work

The research done here can be improved by analysing other games with emergent gameplay and strong communities. In particular, it would be useful to determine if the proposed design patterns are also present in these games. Additional research can be made to extract more design patterns from the gameplay, as well as further examining the existing design patterns that can be related.

For this research paper, more focus was placed on the YouTube community over other Minecraft communities. Further research can expand on the findings done here by analysing the behaviour in different online affinity spaces.

7 Conclusion

In this paper, we examined how some of the mechanics and emergent systems of Minecraft affect the discourse and communities around the game. We analysed how various Minecraft communities interact with the game's systems, grouping our findings into four categories: (1) knowledge sharing and specialised subcommunities, (2) custom game modes, maps and servers, (3) storytelling and content creation, and (4) the continuous evolution of Minecraft. Utilising the MDA framework, we compiled our findings from each category into a set of game design patterns that encourage communities to form and prosper through emergent gameplay. This is the set of game design patterns that emerged from our research:

- Self expression and creativity can be encouraged by using unrestricted, modular components that are open to interpretation
- Systems with hidden complexity and skill allow for mastery and specialisation, leading to players taking separate roles
- Complex systems can be achieved by having multiple components that interact in different ways. These can lead to players making and sharing their own discoveries

- Leaving some of the information in a game obscure can encourage players seeking out information from other players
- Letting players create their own challenges can result in a lot of new content
- Distinct emergent narratives are engaging and sharable, particularly if they change the intensity. Systems can be designed to make it more likely for these stories to occur
- Encouraging self expression and player freedom allows players to tell and share their own stories. Communities can form around these stories, with less focus being placed on the game itself

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